Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S1 (S, FE) S2 (S, FE) Examination December 2024 (2019 Scheme)

Course Code: MAT 101
Course Name: LINEAR ALGEBRA AND CALCULUS
(2019 -Scheme)

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks Marks

1 Find the rank of the matrix $\begin{bmatrix} 1 & 3 & 2 \\ 2 & -1 & 3 \\ 3 & -5 & 4 \\ 1 & 17 & 4 \end{bmatrix}$ (3)

2 What kind of conic section is given by the quadratic form $7x^2 + 6xy + 7y^2 = 200$. (3)

3 Find the slope of the surface $z = xe^{-y} + 5y$ in the y-direction at the point (4,0) (3)

4 Prove that $f_{xy} = f_{yx}$ where $f(x, y) = \ln(x^2 + y^2)$ (3)

5 Find the area bounded by the x-axis, $y = 2x$ *and* $x + y = 1$ using double integrals. (3)

6 Evaluate $\int_0^1 \int_{x^2}^{\sqrt{x}} \int_0^3 dzdydx$ (3)

7 Find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{9}\left(\frac{1}{3}\right)^n$ (3)

8 Determine the rational number representing the decimal number 0.764764764…. (3)

9 Find the Maclaurin series expansion of $f(x) = \ln(1 - x)$ upto 3 terms (3)

10 If $f(x)$ is a periodic function with period $2l$ defined in $[-l, l]$, write the expressions for the Fourier coefficients $a_0, a_n,$ *and* b_n. (3)

PART B

Answer one full question from each module, each question carries 14 marks.

MODULE 1

11 a Using the Gauss elimination method, solve the system of equations $x + 2y + z = 3, 2x + 3y + 2z = 5, 3x - 5y + 5z = 2, 3x + 9y - z = 4$ (7)

b Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$ (7)

12 a Test for consistency and solve the following system of equations: (7)

$2x - y + 3z = 8, -x + 2y + z = 4, 3x + y - 4z = 0$

b Diagonalize the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ (7)

MODULE 2

13 a If $u = f\left({}^{x}/_{y}, {}^{y}/_{z}, {}^{z}/_{x}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$. (7)

b Find relative extrema and saddle points, if any, of the function (7)

$f(x, y) = x^3 + y^3 - 15xy$

14 a If $u = \frac{1}{\sqrt{x^2+y^2+z^2}}$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ (7)

b Find the local linear approximation L to the function $f(x, y, z) = \log{(x + yz)}$ at the point $\text{P}(2,1,-1)$.Compute the error in approximation f by L at the point Q(2.02,0.97,-1.01). (7)

MODULE 3

15 a Evaluate $\iint^{R} ydA$ where R is the region in the first quadrant enclosed between the circle $x^2 + y^2 = 25$ and the line $x + y = 5$. (7)

b Evaluate the integral $\int_0^4 \int_y^4 \frac{x}{x^2+y^2} dxdy$ by first reversing the order of integration. (7)

16 a Find the center of gravity of the triangular lamina with vertices $(0,0), (0,1)\ and\ (1,0)$ and density function $\delta(x, y) = xy$. (7)

b Find the volume bounded by the cylinder $x^2 + y^2 = 4$, the planes $y + z = 3$ and $z = 0$ (7)

MODULE 4

17 a Check the convergence of the series $\frac{3}{4} + \frac{3.4}{4.6} + \frac{3.4.5}{4.6.8} + \frac{3.4.5.6}{4.6.8.10} + \cdots \ldots \ldots$ (7)

b Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}(k+3)}{k(k+1)}$ (ii) $\sum_{k=1}^{\infty} \frac{2}{3^k+5}$ (7)

18	a	Test the convergence of the series (i) $\sum_{k=1}^{\infty}\left(\frac{3k-4}{4k-5}\right)^k$ (ii) $\sum_{k=1}^{\infty}\frac{1}{2k^2+k}$	(7)
	b	Test the absolute or conditional convergence of $\sum_{k=1}^{\infty}\frac{(-1)^{k+1}k^2}{k^3+1}$	(7)

MODULE 5

19	a	Find the Fourier series for $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ \pi, & 0 < x < \pi \end{cases}$ and deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \ldots \ldots.. = \frac{\pi}{4}$	(7)
	b	Find the half range Fourier sine series representation of $f(x) = e^x \; in \; (0,1)$	(7)
20	a	Find the Fourier series for $f(x) = \lvert sinx \rvert, -\pi < x < \pi$	(7)
	b	Obtain the half range Fourier cosine series of $f(x) = x^2$ in $0 < x < 2$	(7)

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